

Question	Marking details	Marks Available					
		AO1	AO2	AO3	Total	Maths	Prac
6	Indicative content Definitions: mesophyte – adapted to conditions of {adequate/moderate/sufficient/owtte} water supply xerophyte – adapted to conditions where water is scarce/groundwater is frozen/named example/dry conditions. hydrophyte – adapted to aquatic conditions/live in water/float on surface of water Pinus/xerophyte Adaptations to reduce transpiration/water (vapour) loss Comments must relate to image Reject adaptations not visible Linked points: • Sunken stomata/stomata in pits + trap humid air/to reduce {diffusion/concentration/water potential} gradient • Small leaves/needle shaped/compact + reduce surface area/SA:vol • {fewer/smaller} stomata + {fewer/smaller} gaps for water loss • <u>Thick</u> cuticle/epidermis + reduce evaporation/water loss • Fibres/sclerenchyma/lignified tissue + provides support/prevents wilting Potamogeton/hydrophyte Linked points: • Cuticle thin/absent + no need to reduce water loss/evaporation • Stomata on {upper/adaxial} surface/no stomata on {lower/abaxial} surface + for gas exchange with air • Air spaces/Aerenchyma/lacunae + provide { buoyancy/flotation} for {light/photosynthesis}/{act as gas reservoir/for gas exchange} • {Little/no} {lignified tissue/xylem} + water provides support • Poorly developed/little xylem + water provided by surroundings	3	6		9		

Question				Marking details	Marks Available					
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				7-9 marks Correct Definitions of xerophytes + hydrophytes + mesophytes. + Full description of Pinus xerophytic adaptations + Full description of Potamageton hydrophytic adaptations The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with few irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately. 4-6 marks Any two from: definitions of mesophytes/xerophytes/hydrophytes description of adaptation of the xerophyte description of adaptation of the hydrophyte The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately. 1-3 marks A definition of xerophyte/hydrophyte/mesophyte or an adaptation of xerophytes or an adaptation of hydrophytes <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
				Question 6 total	3	6		9		